

Radio Frequency Detector

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Here is a circuit that can detect the presence of a radio frequency (RF) signal. If it is close to RF-emitting devices such as a television set or a computer, it shows a high reading in the UV meter, giving us an idea about the intensity of RF radiation in the room. High-intensity RF radiation from mobile phone towers in the vicinity is considered to be harmful by most.

Circuit and working

The circuit diagram of the RF detector is shown in Fig. 1. It is built

around AM/FM detector diodes 1N34 (D1 and D2), an operational amplifier CA3130 (IC1), npn transistor BC547 (T1), connector CON1 for VU meter and a few other components.

Detector diodes D1 and D2 capture the RF from surroundings for the inverting input of IC1. The IC is configured as an inverting amplifier with non-inverting input tied to the negative rail.

Feedback resistor R1 controls the gain of the inverting amplifier. IC CA3130 is a high-gain operational amplifier with sensitive inputs. It is a 15MHz BiMOS operational amplifier with MOSFET inputs and

bipolar output. Its high-speed performance is suitable for low input-current applications.

RF signals are amplified by the inverting amplifier built around IC1. The amplified signal is used to drive the sensitive VU meter. Potmeter VR1 is used to set zero



PARTS LIST

Semiconductors:

- IC1 - CA3130 operational amplifier
- T1 - BC547 npn transistor
- LED1 - 5mm LED
- D1-D2 - 1N34 detector diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-mega-ohm
- R2 - 1-kilo-ohm
- R3 - 470-ohm
- VR1 - 47-kilo-ohm potmeter

Capacitors:

- C1-C2 - 0.22 μ F ceramic disk

Miscellaneous:

- CON1 - 2-pin connector
- BATT.1 - 9V PP3 battery
- VR1 - VU meter
- ANT.1 - 8-pin IC base
- TELESCOPIC ANTENNA

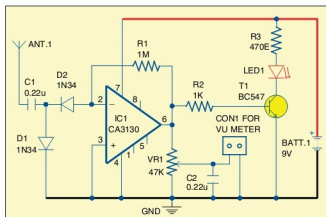


Fig. 1: Circuit diagram of the radio frequency detector

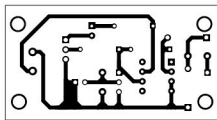


Fig. 2: Actual-size PCB of the radio frequency detector

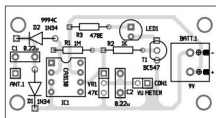


Fig. 3: Component layout of the PCB

level in the meter.

When the circuit gets RF energy from a device or the surroundings,

the meter shows a deflection based on the intensity of RF. It shows full deflection when the unit is close to the RF-generating device. At the same time, transistor T1 conducts and LED1 connected to its collector glows. The glowing of LED1 depends on the intensity of RF; it gives full brightness when RF intensity is very high.

Construction and testing

An actual-size, single-side PCB for the RF detector is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a suitable small box in such a way that potmeter VR1 and VU meter can be fixed on the front side. Ensure proper wiring to avoid any mistake.

Do not solder CA3130 directly on the PCB; instead, use an 8-pin IC base for the IC.

A telescopic antenna is used to increase the sensitivity of the circuit. Use 9V PP3 battery as the power source.

Fig. 4 shows the author's prototype of the RF detector. ●



Fig. 4: Author's prototype



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